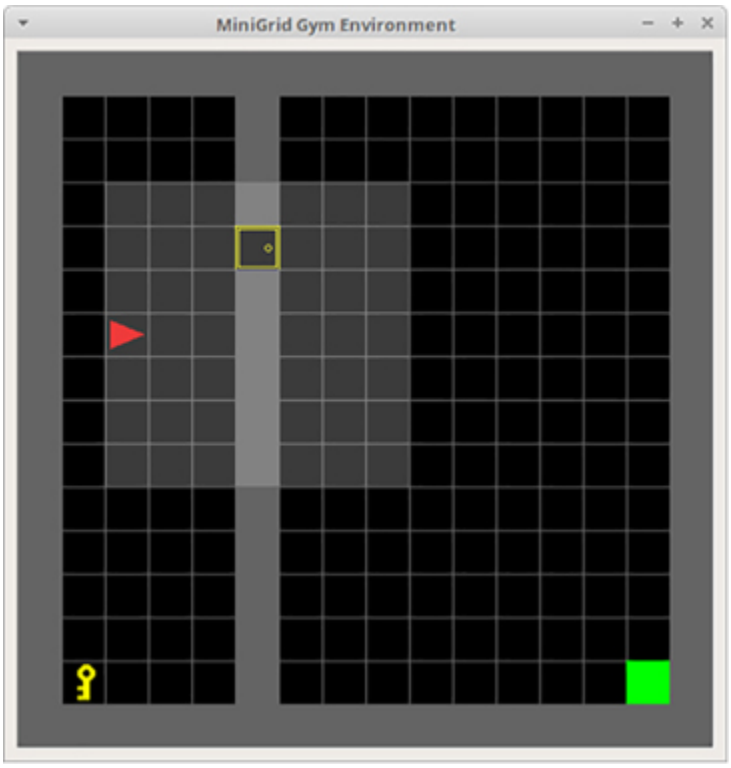



There are a few other complexities to the MiniGrid set of environments. One is that they are partially observable environments, meaning the agent cannot see the whole grid but only a small region immediately surrounding it. Another is that the agent does not simply move left, right, up, and down but has an orientation. The agent can only move forward, turn left, or turn right; it is always oriented in a particular direction and must turn around before moving backward, for example. The agent's partial view of the environment is *egocentric*, meaning the agent sees the grid as if it were facing it. When the agent changes direction without moving position, its view changes. The state we receive from the environment is a $7 \times 7 \times 3$ tensor, so the agent only sees a $7 \times 7$ subregion of the grid in front of it; the last (channel) dimension of the state encodes which object (if any) is present at that position.

This Gridworld environment is a good testbed for our relational module because in order to successfully learn how to play, the agent must learn how to relate the key to the lock and the lock to being able to access the goal, which is all a form of relational reasoning. In addition, the game is naturally represented by a set of objects (or nodes), since each "pixel" position in the